Assignment 6: Railway Fare Calculator

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Calculate train travel fares for multiple passengers using distance base rates, class-based multipliers, age-based concessions, and fixed reservation fees. Process penalties for hypothetical ticket cancellations.

Algorithm

1. Initialize standard travel base variables (distance, rate per km, reservation fee).
2. Use a nested list for passenger records (Name, Age, Class).
3. Iterate through the passengers using a `for` loop.
4. Calculate raw fare using distance and apply a multiplier if traveling 1AC or 2AC.
5. Use `if-elif` statements to apply age concessions (Senior Citizen, Child).
6. Add the standard reservation fee and print the individual ticket fare.

Python Implementation

```
travel_distance = 650 # kilometers
base_rate = 1.5       # Rs per km
reservation_fee = 60  # Flat fee

# Passenger List: [Name, Age, Class]
passengers = [
    ["Aryan", 21, "Sleeper"],
    ["Kavita", 65, "2AC"],
    ["Rohan", 10, "1AC"]
]

print("--- Ticket Booking Summary ---")
for p in passengers:
    name, age, t_class = p

    # Base calculation
    base_fare = travel_distance * base_rate

    # Class multiplier
    if t_class == "1AC":
        base_fare = base_fare * 2.5
    elif t_class == "2AC":
        base_fare = base_fare * 1.5

    # Age-based concessions
    if age >= 60:
        base_fare = base_fare * 0.50 # 50% discount for seniors
    elif age <= 12:
        base_fare = base_fare * 0.40 # 60% discount for children

    final_fare = base_fare + reservation_fee
    print("Passenger:", name, "(Age:", age, ")")
    print("Class:", t_class, "-> Fare: Rs.", round(final_fare, 2))

    # Cancellation context
    penalty = final_fare * 0.20
    print("  * Cancellation Penalty (if cancelled): Rs.", round(penalty, 2))
```